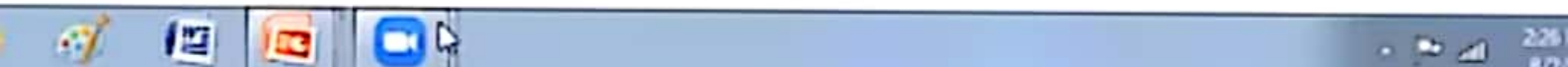


- ① A train takes 23 sec to past completely through a station of 272 m long and 19 sec to another station is 200 m long. The length of the train is
- (a) 140 m (b) 142 m
(c) 144 m (d) 145 m



From the time the front of a train enters a platform, it takes 25 seconds for the back of the train to leave the platform, while travelling at a constant speed of 54 km/h. At the same speed, it takes 14 seconds to pass a man running at 9 km/h in the same directions as the train. What is the length of the train and that of the platform in meters, respectively? (GATE - 18)

- (a) 210 and 140
- (b) 162.5 and 187.5
- (c) 245 and 130
- (d) 175 and 200

(d) 175 m

A train takes 10 seconds to cross a man standing on a platform and 44 seconds to cross the platform. What is the length of the platform if the speed of the train is 72 kmph?

- A) 440 m B) 570 m
C) 680 m D) None of these

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A jogger running at 9 kmph alongside a railway track in 240 metres ahead of the engine of a 120 metres long train running at 45 kmph in the same direction. In how much time will the train pass the jogger

- A) 3.6 sec B) 18 sec
C) 36 sec D) 72 sec

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(2M)

~~GATE~~ – 2020: Two cars start at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is ____.

- (A) 1 (B) 2
(C) 3 (D) 6

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Two trains started at 7AM from the same point. The first train travelled north at a speed of 80k/h and the second train travelled south at a speed of 100km/h. The time at which they were 540 km apart is _____ AM. (GATE - 19)

- (a) 9 (b) 10
(c) 11 (d) 11.30

* Two stations A and B are 110 km apart on a straight line. One train starts from A at 7.00 am and travels towards B at 20 km/h speed. Another train starts from B at 8 am and travels towards A at 25 km/h speed. At what time will they meet?

- (a) 10.30 am (b) 11 am
(c) 10 am (d) none of these

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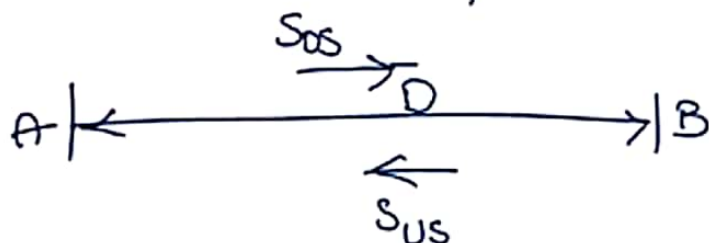
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Two buses start from stations A and B and traveled towards each other at speeds of 50 kmph and 60 kmph respectively. At the time of their meeting the second bus has found 120 km more than first. The distance between A and B is
(a) 1320 km (b) 1330 km
(c) 1310 km (d) 1300 km

Speed of a boat in still water is 16 km/h. If it can travel 20 km downstream in the same time as it can travel 12 km upstream, the rate of stream is

- A) 1 km/h
- B) 2 km/h
- C) 4 km/h
- D) 5 km/h

$$S_B = 20 \text{ km/hr} \quad \left| \quad S_S = 10 \text{ km/hr} \right.$$



$$T_{AB} + T_{BA} = 10 \text{ hrs}$$

$$\frac{D}{20+10} + \frac{D}{20-10} = 10 \Rightarrow$$

② If boat's rate in still water is 20 km/hr and stream rate is 10 km/hr, and it takes him 10 hrs to row to a place and back, then how far is the place?

A) 75 kms B) 60 kms
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C) 70 kms D) 40 kms

$$D \left(\frac{20}{30} \right) = 10 \quad \begin{matrix} T = \frac{\text{Dist.}}{\text{speed}} \\ \text{speed} \end{matrix}$$

$$D = 75 \text{ kms}$$

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3

If Boat's rate in still water is 20 km/hr and it takes him thrice as long to row up as to row down the river. Find the stream rate?

- A) 10 km/hr
- B) 12 km/hr
- C) 5 km/hr
- D) 8 km/hr

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① By selling a calculator for Rs. 418 shopkeeper gains 10%. The cost price of the calculator is

- A) Rs. 459.80
- B) Rs. 408
- C) Rs. 376.20
- D) Rs. 380.00



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③ Sarams purchased 120 reams of paper at Rs. 80 per ream. He spent Rs. 280 on transportation, paid tax at the rate of 40 paise per ream and paid Rs. 72 to the coolie. If he wants to have a gain of 8%, what must be the selling price per ream?

- (a) Rs. 86 (b) Rs. 87.48
(c) Rs. 89 (d) Rs. 90

$Sp = 450$ 10% loss

$Sp = 90\% \text{ CP}$

$450 = \frac{90}{100} \text{ CP} \Rightarrow \text{CP} = 500$

4) By selling an article for Rs. 450, a man loses 10%. The gain or loss percent if he sells it for Rs. 540 is?

- A) Gain 8%
- B) Loss 8%
- C) Gain 20%
- D) Loss 20%

$Sp' = 540$

Gain = 40

Gain % = $\frac{40}{500} \times 100 = 8\%$

A dealer sold a Radio at a loss of 2.5%. Had he sold it for Rs. 100 more, he would have gained 7.5%. To gain 12.5% he should sell it for?

A. Rs 2200 B. Rs 1000 C. Rs 1100 D. Rs 1125



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*×

A fruit seller sold a basket of fruits at 12.5% loss. Had he sold it for Rs. 108 more, he would have made a 10% gain. What is the loss in Rupees incurred by the fruit seller?

(GATE - 2018) → QM

- (a) 48 (b) 52
(c) 60 (d) 108